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MATHEMATICAL LOGIC • COMPUTABILITY THEORY • COARSE GEOMETRY

A Jump Metric on the Arithmetical Degrees:

Rays, a Shadow Tree, Dead Ends, and the Question of Tree-likeness

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Formalisation developed with Claude, an AI system by Anthropic (see Author contributions)

Abstract. For sets $x, y \subseteq \mathbb{N}$ let $\delta(x, y)$ be the least k with $y \leq_T x^{(k)}$, and $d(x, y) = \max(\delta(x, y), \delta(y, x))$. This extended metric on the Turing degrees is finite exactly between mutually arithmetical sets, and we study its component $\mathcal{A}$ of arithmetical degrees. Every canonical ascent is a geodesic ray asymptotic to the trunk $\emptyset, \emptyset', \dots$; these rays determine one ideal point ξ , with Busemann function $-e_\infty$ (eventual high_n level), and the boundary is a singleton iff every geodesic ray has bounded lag. The ultrametric $D(x, y) = \min\{k : x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}\}$ has a ball tree, and $d \leq D \leq d + \max(\varepsilon(x), \varepsilon(y))$ for an excess ε vanishing exactly on the generalised-low degrees. On the sheet spanned by two mutually random generalised-low degrees, d is the longer leg of the tree route, not its length; so d is 2-bi-Lipschitz to a tree metric there yet violates the four-point condition without bound: $(\mathcal{A}, d)$ is not Gromov-hyperbolic; nor is d a graph metric. Excess obeys $e_\infty \leq c_\infty \leq \varepsilon \leq h(x') - 1$. Hidden information is unbounded: generic coding yields first-jump excess E at horofunction level 0, so neither level map into the tree is a quasi-isometry and excess is not Lipschitz; random content hides nothing beyond two rungs. d and D are interleaving distances of the jump flow and of an idempotent canonical flow, the unit-step graph Γ is 3-bi-Lipschitz to d , and D is its minimax edge-weight metric. Tree-likeness is thus Manning's bottleneck property for Γ ; it fails if every countable jump partial order with least element embeds in the degrees preserving $\mathbf{0}$ and the jump: then every finite integer metric embeds isometrically in $(\mathcal{A}, d)$, which is therefore not quasi-isometric to any tree.

Series note. This is Paper A of the series *Rungs, Voids and Effective Closure*, the first programme of the study *Effective Horizons and Cognitive Spaces* (synopsis in [1]), four working papers on a model of agents whose cognitive reach is a Turing ideal and who extend it by uncertified convergence. The core paper [2] supplies the order of reachability (a cognitive space is a Turing ideal, effective closure adds the limits of uncertified approximations, and iterated closure gives the arithmetical ladder); the present paper turns closure steps into a distance and studies its geometry, and can be read independently of the framework; the sequels [3] and [4] add time, populations, finite resources and strategy, and take the resulting dynamics to the Fermi problem.

1 INTRODUCTION

The Turing jump organises the arithmetical degrees into a hierarchy, but the hierarchy is an order, not a distance. This paper studies the metric one obtains by asking, for two degrees x and y , how many jumps each must take before it computes the other, and taking the larger of the two answers. We call it the *rung distance*. It is an

extended metric on all Turing degrees, finite exactly between mutually arithmetical degrees, and we study its principal component, the arithmetical degrees $\mathcal{A}$.

The motivation comes from a model of agents whose cognitive reach is a Turing ideal and who extend that reach by uncertified convergence, one jump per step [2]; in that model $\delta(x, y)$ is the number of such steps an agent at x needs before y is within reach, and d is the lag before two agents become mutually intelligible. None of that motivation is needed here. The object is a natural one in its own right, and the questions it raises are questions of degree theory and coarse geometry: what are its geodesic rays, what is its boundary at infinity, is it hyperbolic, is it tree-like, and is it a graph metric.

Summary of results.

- d is an extended metric; on $\mathcal{A}$ it is sandwiched by height, $|h(x) - h(y)| \leq d(x, y) \leq \max(h(x), h(y))$ where $h(x) = d(x, \emptyset)$ (Section 3).
- Every canonical ascent $x, x', x'', \dots$ is a geodesic ray asymptotic to the trunk $\emptyset, \emptyset', \dots$; all such rays determine one ideal point ξ , identified with true arithmetic, whose Busemann function is $-e_\infty$, the eventual high_n level. Intermediate c.e. degrees give rays that fellow-travel the trunk at distance exactly one for ever without merging into it (Section 4).
- The shadow distance $D(x, y) = \min\{k : x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}\}$ is an ultrametric whose balls form a rooted tree; with the excess $\varepsilon(x)$, the least e with $x^{(k)} \leq_T x \oplus \emptyset^{(k+e)}$ for all k , one has $d \leq D \leq d + \max(\varepsilon(x), \varepsilon(y))$, so families of excess at most e are $3e$ -hyperbolic; ε vanishes exactly on the generalised-low degrees, which sit on the tree exactly (Section 5).
- On the random sheet generated by two mutually random generalised-low degrees a, b of height H and the trunk, $d(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = \max(s, t, H) - \min(s, t)$: the longer leg of the tree route, not its length. Hence d is 2-bi-Lipschitz to the ball tree there (Section 6).
- $(\mathcal{A}, d)$ fails the four-point condition with unbounded defect, so is not Gromov-hyperbolic. Every geodesic ray r has $h(r(i)) = i + O(1)$, so the boundary is a singleton iff every ray has bounded lag $\delta(r(i), \emptyset^{(i)})$. Tree-likeness (Conjecture 1) is reduced to two estimates and shown incompatible with d being a graph metric; d is then shown not to be a graph metric by exhibiting dead ends, pairs a, b at distance $2t \geq 4$ with no neighbour of a closer to b , and the graph distance on the sheet is bounded between $3t - 1$ and $4t - 1$ (Section 7).
- Excess is sandwiched between the horofunction level and the height of the first jump, $e_\infty \leq c_\infty \leq \varepsilon \leq h(x') - 1$, and is *not* decided at the first jump (Friedberg points have first-jump excess 0 and excess E); the horofunction level is 1-Lipschitz. Hidden information is unbounded: coding a generic set with delays drawn from independent generic columns gives, for every E , a point with first-jump excess E and horofunction level 0. So neither the excess-level nor the horofunction-level map into the shadow tree is a quasi-isometry, and excess is not Lipschitz. Conversely, relatively random content is dominated one jump up and hides nothing beyond two rungs; first-jump excess is bounded by the number of rungs needed to dominate the jump, and the bound is attained (Section 7.3).
- Structurally, d and D are the interleaving distances of the jump flow and of the idempotent canonical flow, and excess is the uniform defect between them. The unit-step graph Γ is 3-bi-Lipschitz to d , hence a geodesic model of the same coarse geometry; D is its minimax edge-weight metric, with the balls of the shadow tree as the filtered components; unit edges cross arbitrarily many tree levels; the serial metric $p = \delta(x, y) + \delta(y, x)$ is 2-bi-Lipschitz to d and 0-hyperbolic on the sheet, so the four-point failure is an L^∞/L^1 artefact; and there is exactly one coarse end. Tree-likeness is thereby Manning's bottleneck property for Γ (Section 7.4).
- Under Hypothesis (H), that every countable jump partial order with least element embeds in the degrees preserving $\mathbf{0}$ and the jump, every finite integer metric embeds isometrically in $(\mathcal{A}, d)$, Γ contains isometric cycles of all lengths, and tree-likeness fails (Section 7.5).

The proofs use the limit lemma, the relativised low basis theorem, van Lambalgen's theorem, the jumps of relatively random and relatively generic sets, Solovay tests, and the fact that sets random relative to U' avoid null $\Pi_2^0(U)$ classes.

2 PRELIMINARIES

We use standard notation [18]. Identify $X \subseteq \mathbb{N}$ with its characteristic sequence in Cantor space $2^{\mathbb{N}}$, with metric

$$d(X, Y) = 2^{-\min\{n : X(n) \neq Y(n)\}}, \quad d(X, X) = 0.$$

Convergence $X_s \rightarrow X$ is pointwise stabilisation: for each n , $X_s(n) = X(n)$ for all sufficiently large s .

We write $X \leq_T A$ if $X = \Phi^A$ for some oracle machine Φ . The jump A' is the halting set relative to A ; by the relativised halting theorem, $A' \not\leq_T A$. Iterating from the empty set gives $\emptyset^{(0)} = \emptyset$ and $\emptyset^{(k+1)} = (\emptyset^{(k)})'$, and the ω -jump is the uniform join

$$\emptyset^{(\omega)} = \bigoplus_{k \in \mathbb{N}} \emptyset^{(k)} = \{\langle k, n \rangle : n \in \emptyset^{(k)}\}.$$

By Post's theorem, $X \in \Delta_{k+1}^0$ if and only if $X \leq_T \emptyset^{(k)}$, and X is arithmetical if and only if $X \leq_T \emptyset^{(k)}$ for some k . The set $\emptyset^{(\omega)}$ is not arithmetical and is Turing-equivalent to true arithmetic [14, 19].

A *Turing ideal* is a non-empty class $R \subseteq 2^{\mathbb{N}}$ closed downward under $\leq_T$ and under $\oplus$. For a Turing ideal R let $\text{cl}(R)$ be the class of limits $\lim_s X_s$ of sequences (X_s) whose code $g(n, s) = X_s(n)$ lies in R .

Proposition 1 (Effective closure is one jump). *For a Turing ideal R , $\text{cl}(R) = \{X : X \leq_T A' \text{ for some } A \in R\}$. In particular, for R_0 the computable sets and $R_{k+1} = \text{cl}(R_k)$, $R_k = \{X : X \leq_T \emptyset^{(k)}\}$.*

Proof. This is the relativised limit lemma of Shoenfield [17]: $X \leq_T A'$ if and only if $X = \lim_s g(\cdot, s)$ for some $g \leq_T A$. Downward closure of R gives the first claim, and induction the second. ■

We use three facts about algorithmic randomness, each relative to an oracle Z : the relativised low basis theorem [12], that every non-empty $\Pi_1^0(Z)$ class has a member P with $(P \oplus Z)' \equiv_T Z'$; van Lambalgen's theorem [20], that $a \oplus b$ is Martin-Löf random relative to Z iff a is random relative to Z and b is random relative to $a \oplus Z$; and the fact that if R is random relative to Y' then $(R \oplus Y)' \equiv_T R \oplus Y'$ [13]. We write x is *low_n* if $x^{(n)} \equiv_T \emptyset^{(n)}$ and *high_n* if $x^{(n)} \geq_T \emptyset^{(n+1)}$, for $x \leq_T \emptyset'$.

3 THE RUNG DISTANCE

Throughout, x, y, z range over sets of natural numbers identified with their Turing degrees.

Definition 1 (Rung distance). For sets x, y let $\delta(x, y)$ be the least k with $y \leq_T x^{(k)}$, and $\delta(x, y) = \infty$ if y is not arithmetical in x . Put $d(x, y) = \max(\delta(x, y), \delta(y, x))$.

The quantity $\delta(x, y)$ is the number of closure steps (Proposition 1) an agent at x needs before y lies in its space. It is directed: $\delta(\emptyset^{(k)}, \emptyset) = 0$ while $\delta(\emptyset, \emptyset^{(k)}) = k$.

Lemma 1. $\delta(x, x) = 0$ and $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$. Consequently d is an extended metric on Turing degrees: $d(x, y) = \infty$ exactly when x and y are not arithmetical in each other. Its components are the classes of mutually arithmetical sets, and the component of $\emptyset$ is $\mathcal{A}$.

Proof. If $y \leq_T x^{(j)}$ and $z \leq_T y^{(k)}$ then, since the jump preserves reducibility, $z \leq_T x^{(j+k)}$. The remaining claims are immediate. ■

Write $h(x) = d(x, \emptyset)$, the least k with $x \leq_T \emptyset^{(k)}$.

Lemma 2 (Sandwich). For $x, y \in \mathcal{A}$, $|h(x) - h(y)| \leq d(x, y) \leq \max(h(x), h(y))$.

Proof. Every x satisfies $x^{(k)} \geq_T \emptyset^{(k)}$, so $y \leq_T \emptyset^{(h(y))} \leq_T x^{(h(y))}$ and $\delta(x, y) \leq h(y)$; symmetrically $\delta(y, x) \leq h(x)$. Conversely, if $y \leq_T x^{(k)}$ then $y \leq_T \emptyset^{(h(x)+k)}$, so $h(y) \leq h(x) + \delta(x, y)$. ■

4 A CANONICAL POINT AT INFINITY

Since $(\mathcal{A}, d)$ will turn out not to be hyperbolic (Theorem 2), we fix the boundary notion explicitly. A *geodesic ray* is a map $r : \mathbb{N} \rightarrow \mathcal{A}$ with $d(r(i), r(j)) = |i - j|$. Two rays are *asymptotic* if $\sup_i d(r(i), r'(i)) < \infty$. An *ideal point* is an asymptote class of rays, and $\partial_\infty \mathcal{A}$ is the set of ideal points.

Proposition 2. *For every $x \in \mathcal{A}$ the canonical ascent $x, x', x'', \dots$ is a geodesic ray: $d(x^{(i)}, x^{(j)}) = |i - j|$. Moreover $d(x^{(j)}, \emptyset^{(j)}) = c_j(x)$, where $c_j(x) = h(x^{(j)}) - j$ is non-increasing and bounded by $h(x)$. Hence all canonical ascents in $\mathcal{A}$ are asymptotic to the trunk $\emptyset, \emptyset', \emptyset'', \dots$ and determine a single ideal point $\xi \in \partial_\infty \mathcal{A}$, the asymptote class of the trunk.*

Proof. For $i < j$, $x^{(j)} \leq_T x^{(i+(j-i))}$ and $x^{(j)} \not\leq_T x^{(j-1)}$ by the relativised halting theorem, while $\delta(x^{(j)}, x^{(i)}) = 0$. Next, $\delta(x^{(j)}, \emptyset^{(j)}) = 0$ and $\delta(\emptyset^{(j)}, x^{(j)})$ is the least k with $x^{(j)} \leq_T \emptyset^{(j+k)}$, which is $c_j(x)$. If $x^{(j)} \leq_T \emptyset^{(j+c)}$ then $x^{(j+1)} \leq_T \emptyset^{(j+1+c)}$, so $c_{j+1} \leq c_j$. ■

We identify ξ with true arithmetic. The identification is a correspondence, not an equality: ξ is an asymptote class of rays in $\mathcal{A}$, while $\emptyset^{(\omega)}$ is a set outside $\mathcal{A}$ that is Turing-equivalent to true arithmetic, and is the limit of the sequence $\bigoplus_{k < s} \emptyset^{(k)}$ of arithmetical sets, no single term of which generates the sequence.

Proposition 3 (A horofunction measuring highness). *Let $e_j(x) = \max\{e : \emptyset^{(j+e)} \leq_T x^{(j)}\}$. Then $e_j(x)$ is non-decreasing in j , bounded by $c_j(x)$, and so eventually constant with limit $e_\infty(x)$. The Busemann function of the trunk is*

$$b_\xi(x) = \lim_{m \rightarrow \infty} (d(x, \emptyset^{(m)}) - m) = -e_\infty(x).$$

Proof. Monotonicity: if $\emptyset^{(j+e)} \leq_T x^{(j)}$ then $\emptyset^{(j+1+e)} \leq_T x^{(j+1)}$. For $m \geq h(x)$ we have $\delta(\emptyset^{(m)}, x) = 0$, and $\delta(x, \emptyset^{(m)})$ is the least k with $k + e_k(x) \geq m$. Once e_k has reached e_∞ at some K , for every $m \geq K + e_\infty$ this least k is exactly $m - e_\infty$. ■

The horofunction records the high side of the high/low hierarchy. For $x \leq_T \emptyset'$ and $n \geq 1$, $e_n(x) \geq 1$ exactly when x is high_n , so $e_\infty(x) \geq 1$ exactly when x is high_n for some n . The low side appears instead in the fellow-travelling distance of Proposition 2: $c_n(x) = 0$ exactly when x is low_n . Since $e_\infty(x') = e_\infty(x) + 1$, each jump raises the horofunction level by one, and canonical ascent is vertical motion.

Two asymptotic invariants have now appeared: $c_\infty(x) = \lim_j c_j(x)$, the eventual fellow-travelling distance of the canonical ascent from the trunk, and $e_\infty(x)$, the horofunction level. They need not agree.

Proposition 4 (Fellow-travelling without merging). *For every $x \in \mathcal{A}$, $e_\infty(x) \leq c_\infty(x)$, with equality if and only if $x^{(j)} \equiv_T \emptyset^{(j+e_\infty(x))}$ for all sufficiently large j : the canonical ascent eventually merges, in degree, into a translate of the trunk. Equality fails for every c.e. set x that is neither low_n nor high_n for any n ; such sets exist [18], and for them $c_\infty(x) = 1$ and $e_\infty(x) = 0$.*

Proof. $e_j \leq c_j$ was noted above, and $\emptyset^{(j+e_\infty)} \leq_T x^{(j)} \leq_T \emptyset^{(j+c_\infty)}$ for large j gives the characterisation of equality. For x c.e. and intermediate, $\emptyset^{(j)} <_T x^{(j)} <_T \emptyset^{(j+1)}$ for every $j \geq 1$: the first inequality is strict because x is not low_j , the second because x is not high_j . Hence $h(x^{(j)}) = j + 1$ and $\emptyset^{(j+1)} \not\leq_T x^{(j)}$. ■

So a canonical ascent is always asymptotic to the trunk (Proposition 2) but need not merge into it: an intermediate agent climbs beside the trunk at distance exactly one for ever, at every stage strictly above the canonical content of its rung and strictly below that of the next. Its ray still determines the same ideal point ξ , since asymptotic rays are identified.

5 THE SHADOW TREE

Definition 2 (Excess and shadow distance). The *excess* $\varepsilon(x)$ of $x \in \mathcal{A}$ is the least e such that $x^{(k)} \leq_T x \oplus \emptyset^{(k+e)}$ for every k ; it exists and $\varepsilon(x) \leq h(x)$. The *shadow distance* is $\rho(x, y) = \min\{k : y \leq_T x \oplus \emptyset^{(k)}\}$, and $D(x, y) = \max(\rho(x, y), \rho(y, x))$.

An agent of excess 0 is *generalised-low*: $x^{(k)} \equiv_T x \oplus \emptyset^{(k)}$ for all k . Its jumps contain nothing beyond itself and the canonical content $\emptyset^{(k)}$.

Theorem 1 (Shadow Tree Theorem; Rook's Tree). **1.** D is an ultrametric on $\mathcal{A}$: $D(x, y) \leq k$ if and only if $x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}$.

2. For all $x, y \in \mathcal{A}$, $d(x, y) \leq D(x, y) \leq d(x, y) + \max(\varepsilon(x), \varepsilon(y))$.

3. Any family of agents of excess at most e is $3e$ -hyperbolic in the sense of Gromov.

Proof. (1) $D(x, y) \leq k$ says $y \leq_T x \oplus \emptyset^{(k)}$ and $x \leq_T y \oplus \emptyset^{(k)}$, which is equivalent to $x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}$. Call this relation E_k . Each E_k is an equivalence relation, and $E_k \subseteq E_{k+1}$ (join both sides with $\emptyset^{(k+1)}$). So $D(x, y) = \min\{k : x E_k y\}$, and a metric defined by a nested family of equivalence relations is an ultrametric. (2) Since $x \oplus \emptyset^{(k)} \leq_T x^{(k)}$, $\delta \leq \rho$. If $y \leq_T x^{(k)}$ then $y \leq_T x \oplus \emptyset^{(k+\varepsilon(x))}$, so $\rho \leq \delta + \varepsilon(x)$. Symmetrise. (3) By (2), each Gromov product computed with d differs by at most $3e/2$ from the one computed with D , which satisfies the four-point condition exactly. Hence d satisfies it with constant $3e$. ■

Corollary 1 (The ball tree, and generalised-low agents on it). The E_k -classes for $k \in \mathbb{N}$ form a rooted tree $\mathcal{T}$ under inclusion, the ball tree of D : the vertex $[x]_k$ at level k is the class of x under E_k , its parent is $[x]_{k+1}$, and every vertex at level $h(x)$ or above on the branch of x is the trunk vertex $[\emptyset]_k$. The vertex $[x]_k$ may be labelled by the degree $x \oplus \emptyset^{(k)}$, which determines it. On generalised-low agents $d = D$, so d is an ultrametric there and generalised-low agents are exactly the level-0 leaves at which d and D agree; two such agents are at distance k exactly when supplying both with the canonical content $\emptyset^{(k)}$ makes them Turing-equivalent.

The labels are not themselves generalised-low: $\varepsilon(x \oplus \emptyset^{(k)}) = k$ for a generalised-low x with $h(x) > k$, since $(x \oplus \emptyset^{(k)})^{(j)} \equiv_T x \oplus \emptyset^{(k+j)}$ while $(x \oplus \emptyset^{(k)}) \oplus \emptyset^{(j+e)} \equiv_T x \oplus \emptyset^{(\max(k, j+e))}$. Accordingly d and D differ on the labels ($d(x \oplus \emptyset^{(k)}, x \oplus \emptyset^{(k+1)}) = 1$ while $D = k + 1$), and the tree is a tree of balls, not a subspace of $(\mathcal{A}, d)$. This is exactly the source of the excess t carried by the points m_a, m_b in Theorem 2 below.

This gives a non-Archimedean tree of nested balls, structurally analogous to the 2-adic picture: the balls of D are nested or disjoint, as in $\mathbb{Z}_2$, and the tree resembles the Bruhat-Tits tree drawn with a distinguished end at the top. It is not that tree: its branching is infinite and no group action is claimed.

6 THE RANDOM SHEET: A CLOSED FORM

The shadow tree controls every distance up to excess, and one might hope that excess measured relative to an agent's own base on the trunk (so that trunk points count as excess-free) suffices for hyperbolicity. It does not. To see exactly how it fails we compute d on a family where everything is explicit.

Fix $H \geq 1$ and let a, b be sets with the following properties, which Lemma 3 below provides: each is generalised-low with $h(a) = h(b) = H$; $\emptyset^{(k+1)} \not\leq_T a \oplus \emptyset^{(k)}$ and likewise for b , for every k (so $e_k(a) = e_k(b) = 0$); and $\rho(a, b) = \rho(b, a) = H$. Call the family $\{a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)} : s, t \in \mathbb{N}\}$ the *random sheet* over (a, b) .

Proposition 5 (Closed form on the random sheet). For all $s, t \in \mathbb{N}$, writing $M = \max(s, t, H)$,

$$d(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = M - \min(s, t),$$

$$D(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = M \quad (\text{unless } s = t \geq H, \text{ when the two points coincide}),$$

and on a single branch $d(a \oplus \emptyset^{(s)}, a \oplus \emptyset^{(t)}) = |s - t|$, $D = \max(s, t)$ for $s \neq t$. Moreover $\varepsilon(a \oplus \emptyset^{(s)}) = s$ and $h(a \oplus \emptyset^{(s)}) = \max(s, H)$.

Proof. Put $p = a \oplus \emptyset^{(s)}$, $q = b \oplus \emptyset^{(t)}$. Generalised lowness gives $p^{(k)} \equiv_T a \oplus \emptyset^{(s+k)}$. Now $b \leq_T a \oplus \emptyset^{(j)}$ if and only if $j \geq H$, and $\emptyset^{(t)} \leq_T a \oplus \emptyset^{(j)}$ if and only if $j \geq t$ (because $e_j(a) = 0$). So $q \leq_T p^{(k)}$ if and only if $s+k \geq \max(H, t)$, giving $\delta(p, q) = \max(0, \max(H, t) - s)$ and symmetrically $\delta(q, p) = \max(0, \max(H, s) - t)$; the maximum of the two is $M - \min(s, t)$. For D , $q \leq_T p \oplus \emptyset^{(k)} = a \oplus \emptyset^{(\max(s, k))}$ if and only if $\max(s, k) \geq \max(H, t)$, so $\rho(p, q) = \max(H, t)$ when $\max(H, t) > s$ and 0 otherwise; symmetrising gives M except in the stated degenerate case. The single-branch and excess computations are the same calculation with b replaced by a . ■

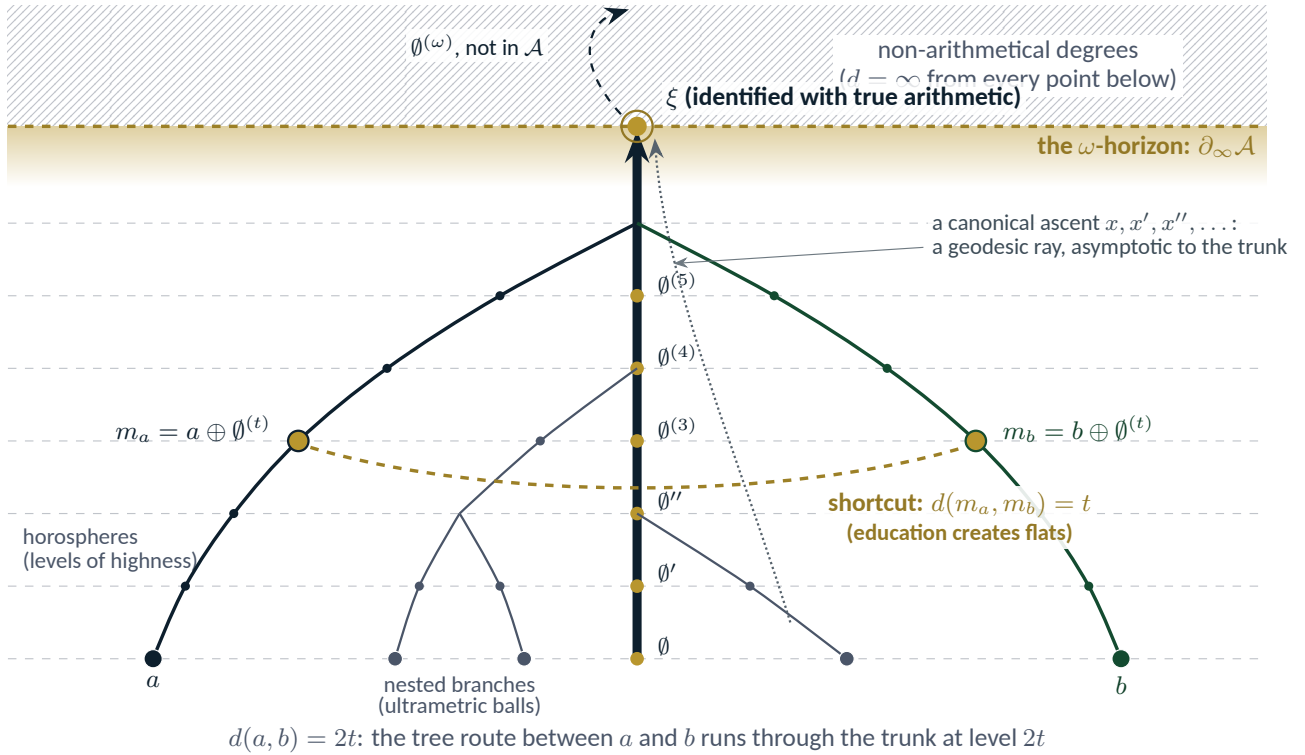


Figure 1: Rook’s Tree (schematic; horizontal positions are illustrative). The trunk $\emptyset, \emptyset', \emptyset'', \dots$ rises towards the ideal point ξ , identified with true arithmetic, on the ω -horizon $\partial_\infty \mathcal{A}$. Dashed horizontal lines are horospheres centred at ξ , measuring highness. Branches are chains of D -balls (Corollary 1): the vertex on the branch of x at level k is the ball $[x]_k$, labelled by $x \oplus \emptyset^{(k)}$, and each branch joins the trunk at height $h(x)$; generalised-low agents are leaves on which d agrees with the tree. Branches that merge below the trunk form nested ultrametric balls. The gold points m_a, m_b are two random agents after absorbing the canonical content $\emptyset^{(t)}$: they sit at distance t , although the route through the tree would place them $2t$ apart, which is the flat of Theorem 2. Above the horizon lie the non-arithmetical degrees, at infinite rung distance from every point of the figure.

Proof. Let $Z = \emptyset^{(H-1)}$. We use three standard facts relative to Z : the relativised low basis theorem [12]; van Lambalgen’s theorem [20]; and the fact that a set R random relative to Y' satisfies $(R \oplus Y)' \equiv_T R \oplus Y'$ [13]. The class of sets Martin-Löf random relative to Z contains a non-empty $\Pi_1^0(Z)$ class, so by the low basis theorem it has a member P with $(P \oplus Z)' \equiv_T \emptyset^{(H)}$. Split $P = a \oplus b$. By van Lambalgen, a and b are each random relative to Z and random relative to each other over Z . Iterating the third fact gives $a^{(k)} \equiv_T a \oplus \emptyset^{(k)}$ for $k \leq H - 1$, and lowness gives $a^{(k)} \equiv_T \emptyset^{(k)}$ for $k \geq H$; so a is generalised-low and $h(a) = H$. If $a \oplus \emptyset^{(k)}$ computed $\emptyset^{(k+1)}$ for some k , then every later jump would stay one level ahead of the trunk, contradicting $a^{(H)} \equiv_T \emptyset^{(H)}$. By van Lambalgen, $b \not\leq_T a \oplus Z$, so $\rho(a, b) = \rho(b, a) = H$. The same holds for b . $\blacksquare$

Corollary 2 (The sheet is a tree up to a factor of two). *Let $\mathcal{T}$ be the ball tree with its path metric $d_{\mathcal{T}}$, and map each point x of the random sheet to the vertex $[x]_{\varepsilon(x)}$. On the sheet this map is injective on degrees and satisfies $d \leq d_{\mathcal{T}} \leq 2d$. Concretely, the tree route from $a \oplus \emptyset^{(s)}$ to $b \oplus \emptyset^{(t)}$ climbs $M - s$ to the meet at height M and descends $M - t$; the rung distance is the longer leg of this route, $M - \min(s, t)$, and the path metric is their sum.*

Proof. By Proposition 5, $\varepsilon(a \oplus \emptyset^{(s)}) = s$, so the image vertex sits at height s , and the meet of the images is at height $M = D$. Then $d_{\mathcal{T}} = (M - s) + (M - t)$ while $d = M - \min(s, t)$, and $\max \leq \text{sum} \leq 2 \max$. $\blacksquare$

7 NON-HYPERBOLICITY, TREE-LIKENESS AND A DICHOTOMY

Because d takes the longer leg rather than the sum, it is not a tree metric, and the defect is unbounded.

Theorem 2. $(\mathcal{A}, d)$ is not Gromov-hyperbolic.

Proof. Fix $H = 2t$ and take a, b from Lemma 3. Put $m_a = a \oplus \emptyset^{(t)}$ and $m_b = b \oplus \emptyset^{(t)}$. Proposition 5 gives

$$d(a, m_a) = d(b, m_b) = t, \quad d(m_a, m_b) = t, \quad d(a, b) = d(a, m_b) = d(b, m_a) = 2t.$$

The three pairwise sums are $d(a, b) + d(m_a, m_b) = 3t$, $d(a, m_a) + d(b, m_b) = 2t$ and $d(a, m_b) + d(b, m_a) = 4t$. The four-point condition requires the largest to exceed the second largest by at most 2δ , forcing $\delta \geq t/2 = H/4$. Since H is arbitrary, no δ works. ■

Remark 1 (Reading the obstruction). The four agents are not exotic: two mutually algorithmically random, generalised-low agents, and the same two agents after each has absorbed the canonical content $\emptyset^{(t)}$. Absorbing shared canonical knowledge halves their separation, since $d(m_a, m_b) = t$ while $d(a, b) = 2t$, a lateral shortcut that no tree admits. Each fixed base level is ultrametric (Corollary 1), and non-hyperbolicity comes from mixing base levels. In this sense *education creates flats*. The bound of Theorem 1(3) is of the right order: m_a and m_b have excess t , and the defect they produce is linear in t .

The theorem should nevertheless be read with Corollary 2 beside it. $(\mathcal{A}, d)$ is integer-valued and has no midpoints, so it is not a geodesic space, and off geodesic spaces the four-point condition is not a quasi-isometry invariant. The same quadruple, measured in the path metric of the ball tree, has distances $t, t, 2t, 3t, 3t, 4t$ with pairwise sums $6t, 2t, 6t$: it is 0-hyperbolic there. What Theorem 2 shows is that d is the *max-leg* of the tree route and not its length; both are cognitively meaningful and they disagree. The rung distance is the mutual-intelligibility lag (the climb the further party must make); the path metric is the cost of sequential traversal. “Education creates flats” is a statement about the first, and is false of the second.

Conjecture 1 (Tree-likeness). *There is a tree T and a quasi-isometry $(\mathcal{A}, d) \rightarrow T$; equivalently (Theorem 5), the unit-step graph Γ has Manning’s bottleneck property. Expected false: Theorem 7 refutes it under a hypothesis on jump partial orders.*

The natural candidate is the ball tree with the excess-level map $\Phi(x) = [x]_{\varepsilon(x)}$. That candidate, and its horofunction-level variant (Conjecture 2), are both refuted in Section 7.3; the conjecture is left as a bare existence statement. For the record, here is what the excess-level map would have needed.

Write $u = \varepsilon(x)$, $v = \varepsilon(y)$ and $M = \max(u, v, D(x, y))$, the height of the meet of $\Phi(x)$ and $\Phi(y)$, so that $d_{\mathcal{T}}(\Phi x, \Phi y) = (M - u) + (M - v)$. Two estimates are available at once. By Theorem 1(2), $M \leq d + \max(u, v)$, whence

$$d_{\mathcal{T}}(\Phi x, \Phi y) \leq 2d(x, y) + |\varepsilon(x) - \varepsilon(y)|;$$

and since $d \leq D \leq M$ and $d_{\mathcal{T}} \geq M - \min(u, v)$,

$$d(x, y) \leq d_{\mathcal{T}}(\Phi x, \Phi y) + \min(\varepsilon(x), \varepsilon(y)).$$

So Φ is a quasi-isometry if and only if two estimates hold: whether excess is Lipschitz in rung distance, $|\varepsilon(x) - \varepsilon(y)| \leq C d(x, y) + c$, and whether the additive $\min(\varepsilon)$ in the second bound can be traded for a multiplicative constant, that is, whether $d(x, y) \leq C(D(x, y) - \min(\varepsilon(x), \varepsilon(y))) + c$. Both hold with $C = 1$ and $c = 0$ on the random sheet and on the trunk. The first fails in general (Corollary 5); the conjecture is restated in Conjecture 2 with the horofunction level, for which the Lipschitz half is free, and that too fails (Theorem 4). The naive sharpening of Theorem 1(2) that would settle the second at a stroke, $D(x, y) - \min(\varepsilon(x), \varepsilon(y)) \leq d(x, y)$, is *false*: for b as in the proof above, $d(b, \emptyset^{(t)}) = t$ while $D(b, \emptyset^{(t)}) = 2t$ and $\varepsilon(b) = 0$. So Theorem 1(2) is tight as stated, and the map, if it worked, would need the multiplicative slack. If the conjecture holds, the headline of this section becomes: *the arithmetical degrees under rung distance are quasi-isometric to a tree*, which says considerably more than non-hyperbolicity.

Proposition 6 (Every geodesic ray climbs at the trunk’s rate). *Let r be a geodesic ray in $(\mathcal{A}, d)$. Then:*

1. $h(r(i)) - i$ is non-increasing, and for $i > h(r(0))$ it lies in $[0, h(r(0))]$; hence it is eventually equal to a constant $c(r)$, the height defect of the ray.
2. Excess is unbounded along r : $\max_{j \leq i} \varepsilon(r(j)) \geq i - 1$ for every i .

3. For all large i , $d(r(i), \emptyset^{(i)}) = \max(\delta(r(i), \emptyset^{(i)}), c(r))$. Consequently r is asymptotic to the trunk, and so represents ξ , if and only if the lag $\delta(r(i), \emptyset^{(i)})$ is bounded in i .

Proof. (1) By Lemma 2, $i = d(r(0), r(i)) \leq \max(h(r(0)), h(r(i)))$, so $h(r(i)) \geq i$ once $i > h(r(0))$; and $h(r(i+1)) \leq h(r(i)) + d(r(i), r(i+1)) = h(r(i)) + 1$. A non-increasing integer sequence bounded below is eventually constant, and the bound $c(r) \leq h(r(0))$ is the value at $i = 0$. (2) $D(r(0), r(i)) \geq d(r(0), r(i)) = i$, while by ultrametricity and Theorem 1(2), $D(r(0), r(i)) \leq \max_{j < i} D(r(j), r(j+1)) \leq 1 + \max_{j \leq i} \varepsilon(r(j))$. (3) $\delta(\emptyset^{(i)}, r(i))$ is the least k with $r(i) \leq_T \emptyset^{(i+k)}$, that is $\max(0, h(r(i)) - i)$, which equals $c(r)$ for large i by (1). ■

Part (1) says that no ray can outrun the trunk in height: every ray has $h(r(i)) = i + O(1)$, so the “horosphere” level of a ray is fixed up to a bounded defect. Part (3) then reduces the question of the boundary to a question about *lag* rather than height. A second ideal point exists if and only if some geodesic ray r has $\delta(r(i), \emptyset^{(i)})$ unbounded: the ray keeps pace with the trunk in height, but its own jumps recover the canonical content $\emptyset^{(i)}$ of its level only after an unboundedly long climb. Single points with large lag are plentiful (for a as in Lemma 3, with $h(a) = H$ and $e_k(a) = 0$ for all k , one computes $\delta(a, \emptyset^{(i)}) = i$ and $\delta(\emptyset^{(i)}, a) = \max(0, H - i)$, so $d(a, \emptyset^{(i)}) = \max(i, H - i)$ and a sits at distance $\lceil H/2 \rceil$ from the whole trunk); what is not known is whether such points can be strung along a ray. Canonical ascents and the branches $a \oplus \emptyset^{(i)}$ of the random sheet both have bounded lag.

Remark 2 (Two further questions, and a dichotomy). (i) *Is $\partial_\infty \mathcal{A}$ a single point?* By Proposition 6(3) this is the question whether every geodesic ray has bounded lag. We conjecture that it does, but the evidence is thin: it amounts to the two families of rays we can compute.

(ii) *Is d a graph metric?* Let Γ be the graph on $\mathcal{A}$ with an edge between x and y when $d(x, y) = 1$, and d_Γ its path metric; always $d \leq d_\Gamma$. Γ is not the ball tree: on the random sheet, $a \oplus \emptyset^{(H-1)}$ and $b \oplus \emptyset^{(H-1)}$ are adjacent in Γ while their tree distance is 2. Canonical ascents show $d_\Gamma = d$ on pairs $(x, x^{(n)})$; Theorem 3 below shows it fails for a, b of Lemma 3 as soon as $H \geq 4$.

The two questions are linked to Conjecture 1 by a dichotomy. If $d_\Gamma = d$, then $(\mathcal{A}, d)$ is a geodesic metric space in the discrete sense, and for geodesic spaces the four-point condition is equivalent to Gromov hyperbolicity and is a quasi-isometry invariant; Theorem 2 then shows that $(\mathcal{A}, d)$ is not quasi-isometric to any tree, and Conjecture 1 is false. Conversely, if Conjecture 1 holds, then $d_\Gamma \neq d$, and some pair at finite distance admits no unit-step chain of that length. At most one of “ d is a graph metric” and “ d is quasi-isometric to a tree” can hold, and it is possible that neither does. The next subsection shows that the first does not; Section 7.4 then shows that d is nevertheless 3-bi-Lipschitz to d_Γ , so that Γ is the geodesic model on which the second must be decided, and that the four-point failure of Theorem 2 is an artefact of the non-geodesic metric which does not bear on it.

7.1 The rung distance is not a graph metric

The dichotomy is settled on its first branch: d is not a graph metric. The proof uses two “no common information” lemmas, both resting on one standard fact.

Lemma 4. *A set that is Martin-Löf random relative to U' lies in no null $\Pi_2^0(U)$ class.*

Proof. Let $P = \bigcap_n V_n$ with V_n nested $\Sigma_1^0(U)$ open sets and $\mu(P) = 0$. Each $\mu(V_n)$ is left-c.e. relative to U and hence computable from U' , so U' can find, for each k , an n_k with $\mu(V_{n_k}) < 2^{-k}$. Then $(V_{n_k})_k$ is a U' -Martin-Löf test covering P . ■

Lemma 5 (No common information, I). *Let a be random relative to U' . If $z \leq_T a \oplus U$ and $z \leq_T U'$ then $z \leq_T U$.*

Proof. Suppose $z \not\leq_T U$. By the theorem of Sacks relativised to U [15], the class $\{A : z \leq_T A \oplus U\}$ is null, so each $C_e = \{A : \Phi_e^{A \oplus U} = z\}$ is null. Since $z \leq_T U'$, each predicate “ $z(n) = v$ ” is $\Delta_2^0(U)$, so C_e , which says that $\Phi_e^{A \oplus U}$ is total and that for all n, s, v , if $\Phi_e^{A \oplus U}(n)[s] \downarrow = v$ then $z(n) = v$, is a $\Pi_2^0(U)$ class. By Lemma 4, $a \notin C_e$ for every e , so $z \not\leq_T a \oplus U$. ■

Lemma 6 (No common information, II). *Let a be random relative to V' and let $W \leq_T V$. If $z \leq_T a \oplus W$ and $z \leq_T V$ then $z \leq_T W$.*

Proof. As before with W in place of U : if $z \not\leq_T W$ then each $\{A : \Phi_e^{A \oplus W} = z\}$ is null, and it is Π_2^0 relative to $z \oplus W$, hence $\Pi_2^0(V)$; apply Lemma 4. ■

Theorem 3 (Dead ends). *Let $t \geq 2$, $H = 2t$, and let a, b be as in Lemma 3. Then $d(a, b) = 2t$, but every z with $d(a, z) = 1$ satisfies $d(z, b) \geq 2t$. In particular no chain of $2t$ unit steps joins a to b , and d is not a graph metric.*

Proof. Let $d(a, z) = 1$ and suppose $d(z, b) \leq 2t - 1$. Then $z \leq_T a' \equiv_T a \oplus \emptyset'$, $a \leq_T z'$, and $z \leq_T b^{(2t-1)} \equiv_T b \oplus \emptyset^{(2t-1)}$. Put $U = b \oplus \emptyset^{(2t-2)}$; since b is generalised-low, $U' \equiv_T b \oplus \emptyset^{(2t-1)}$, and a is random relative to U' by van Lambalgen's theorem. Lemma 5 gives $z \leq_T U$. Now put $V = U$ and $W = \emptyset'$; a is random relative to V' , $z \leq_T a \oplus W$ and $z \leq_T V$, so Lemma 6 gives $z \leq_T \emptyset'$. Hence $a \leq_T z' \leq_T \emptyset''$, contradicting $h(a) = 2t \geq 4$. ■

At $t = 1$ the chain exists ($d(a, \emptyset') = d(\emptyset', b) = 1$), so the obstruction begins exactly at $H = 4$. The same lemmas bound the graph distance from below.

Corollary 3. *With a, b, t as in Theorem 3, $3t - 1 \leq d_\Gamma(a, b) \leq 4t - 1$.*

Proof. Upper bound: the route $a, a \oplus \emptyset', \dots, a \oplus \emptyset^{(2t-1)}, b \oplus \emptyset^{(2t-1)}, \dots, b$ has unit steps by Proposition 5 and length $4t - 1$. Lower bound: let $a = z_0, \dots, z_m = b$ be a Γ -path. Inductively $z_i \leq_T a \oplus \emptyset^{(i)}$ and $z_i \leq_T b \oplus \emptyset^{(m-i)}$. Take $i = t - 1$ and suppose $m \leq 3t - 2$, so that $i \leq m - i \leq 2t - 1$. If $m - i \leq 2t - 2$, Lemma 6 with $V = b \oplus \emptyset^{(m-i)}$ and $W = \emptyset^{(i)}$ gives $z_i \leq_T \emptyset^{(i)}$; if $m - i = 2t - 1$, first apply Lemma 5 with $U = b \oplus \emptyset^{(2t-2)}$ and then Lemma 6. Either way $z_i \leq_T \emptyset^{(t-1)}$, and since $d(a, z_i) \leq i$ we get $a \leq_T z_i^{(t-1)} \leq_T \emptyset^{(2t-2)}$, contradicting $h(a) = 2t$. ■

Remark 3. Theorem 3 closes the first branch of the dichotomy in Remark 2: since d is not a graph metric, the four-point failure of Theorem 2 is not an obstruction to tree-likeness, and Conjecture 1 stands. It also identifies the geodesic space that the coarse-geometric questions should be asked of: the graph Γ . If $d_\Gamma \leq C d + c$ (true on the sheet with $C = 2$), then $(\mathcal{A}, d)$ and Γ are quasi-isometric and Conjecture 1 is equivalent to the Gromov-hyperbolicity of Γ together with the quasi-isometry of Γ to a tree. Theorem 5 below proves this bound with $C = 3, c = 0$.

7.2 The excess sandwich; the horofunction level is Lipschitz

The first of the two estimates asks whether excess is Lipschitz. We first locate excess among the invariants already in play.

Lemma 7 (Excess sandwich). *For $x \in \mathcal{A}$ write $e_1^*(x) = \rho(x, x') - 1 = \min\{e : x' \leq_T x \oplus \emptyset^{(e+1)}\}$ for the first-jump excess. Then*

$$e_1^*(x) \leq \varepsilon(x) \quad \text{and} \quad e_\infty(x) \leq c_\infty(x) \leq \varepsilon(x) \leq c_1(x) = h(x') - 1 \leq h(x).$$

In particular $\varepsilon(x) = 0$ forces $e_\infty(x) = 0$, and $\varepsilon(\emptyset^{(n)}) = n$.

Proof. $e_1^* \leq \varepsilon$ is the case $k = 1$ of the definition. For $\varepsilon \leq c_1$: c_j is non-increasing in j (if $x^{(j)} \leq_T \emptyset^{(j+c)}$ then $x^{(j+1)} \leq_T \emptyset^{(j+1+c)}$), and $x^{(k)} \leq_T \emptyset^{(k+c_k)} \leq_T x \oplus \emptyset^{(k+c_k)}$, so the least e with $x^{(k)} \leq_T x \oplus \emptyset^{(k+e)}$ is at most $c_k \leq c_1$; and $c_1 = h(x') - 1$ by definition. For $c_\infty \leq \varepsilon$: choose j with $j + \varepsilon \geq h(x)$, so that $x \oplus \emptyset^{(j+\varepsilon)} \equiv_T \emptyset^{(j+\varepsilon)}$; then $x^{(j)} \leq_T \emptyset^{(j+\varepsilon)}$ for all such j , whence $c_j \leq \varepsilon$ eventually. $e_\infty \leq c_\infty$ is Proposition 4. For the trunk, $(\emptyset^{(n)})^{(k)} = \emptyset^{(n+k)} \leq_T \emptyset^{(n)} \oplus \emptyset^{(k+e)} \equiv_T \emptyset^{(\max(n, k+e))}$ if and only if $e \geq n$ (take $k \geq 1$). ■

Remark 4 (Excess is not decided at the first jump). The natural guess $\varepsilon = e_1^*$ is false, and the gap is unbounded. Let x be a Friedberg point, $x' \equiv_T x \oplus \emptyset' \equiv_T \emptyset^{(E+1)}$ [10]. Then $e_1^*(x) = 0$, but $x'' \equiv_T \emptyset^{(E+2)}$ while $x \oplus \emptyset^{(2+e)} \leq_T \emptyset^{(\max(E+1, 2+e))}$, so $\varepsilon(x) = E$; indeed $\varepsilon(x) = e_\infty(x) = c_\infty(x) = E$, the whole excess being highness. The tempting argument for equality uses $(A \oplus B)' \leq_T A' \oplus B'$, which is false in general: the jump of a join can strictly exceed the join of the jumps, and the Friedberg point with $A = x, B = \emptyset'$ is a witness.

An earlier draft of this paper asserted equality and drew from it a non-Lipschitz theorem for excess; both are withdrawn; the Lipschitz question is settled negatively, by a different route, in Section 7.3.

The computed cases are consistent with excess being close to the horofunction level: on the trunk and on Friedberg points $\varepsilon = e_\infty$; on the random sheet $\varepsilon(a \oplus \emptyset^{(s)}) = s = e_\infty(a \oplus \emptyset^{(s)})$ (Corollary 1 and the discussion following it); on intermediate c.e. degrees $\varepsilon = 1$ and $e_\infty = 0$, since such x is not low_k for any k while $x^{(k)} \leq_T \emptyset^{(k+1)}$. Every example has $\varepsilon - e_\infty \in \{0, 1\}$.

Proposition 7 (The horofunction level is 1-Lipschitz). $|e_\infty(x) - e_\infty(y)| \leq d(x, y)$ for all $x, y \in \mathcal{A}$.

Proof. $b_\xi = -e_\infty$ is a Busemann function of $(\mathcal{A}, d)$, hence 1-Lipschitz. Directly: if $x \leq_T y^{(n)}$ then $y^{(n+k)} \geq_T x^{(k)} \geq_T \emptyset^{(k+e_k(x))}$, so $e_{n+k}(y) \geq e_k(x) - n$; let $k \rightarrow \infty$ and symmetrise. ■

Proposition 8 (Bounded hidden information makes excess Lipschitz). If $\varepsilon \leq e_\infty + K$ on $\mathcal{A}$ then $|\varepsilon(x) - \varepsilon(y)| \leq d(x, y) + 2K$ for all x, y .

Proof. $e_\infty \leq \varepsilon \leq e_\infty + K$ (Lemma 7) and Proposition 7. ■

So the Lipschitz estimate for the excess-level map would follow from a bound on $\varepsilon - e_\infty$. Section 7.3 shows that no such bound exists, and that the Lipschitz estimate fails.

Conjecture 2 (Tree-likeness, revised; *refuted in Section 7.3*). $\Phi^*(x) = [x]_{e_\infty(x)}$ is a quasi-isometry from $(\mathcal{A}, d)$ to the ball tree.

Write $u = e_\infty(x)$, $v = e_\infty(y)$, $M = \max(u, v, D(x, y))$, so $d_{\mathcal{T}}(\Phi^*x, \Phi^*y) = 2M - u - v$, and note $M - \min(u, v) = \max(|u - v|, D(x, y) - \min(u, v))$ and $M - \min(u, v) \leq d_{\mathcal{T}} \leq 2(M - \min(u, v))$. By Proposition 7 the term $|u - v|$ is at most d . Hence Conjecture 2 is equivalent to the single two-sided estimate

$$d(x, y) \asymp \max(|e_\infty(x) - e_\infty(y)|, D(x, y) - \min(e_\infty(x), e_\infty(y)))$$

up to multiplicative and additive constants. It holds with equality on the random sheet, where $D = \max(s, t, H)$ and $e_\infty(a \oplus \emptyset^{(s)}) = s$, so the right-hand side is exactly $\max(s, t, H) - \min(s, t) = d$; on the trunk; on Friedberg points $(x' \equiv_T x \oplus \emptyset' \equiv_T \emptyset^{(E+1)}: d(x, \emptyset^{(E+1)}) = 1 = D, e_\infty(x) = E)$; and on intermediate c.e. degrees against $\emptyset$ and $\emptyset'$.

Proposition 9 (A necessary condition). If Conjecture 2 holds then $e_1^*(x) - e_\infty(x)$ is bounded on $\mathcal{A}$; since $e_1^* \leq \varepsilon$, this is implied by, but weaker than, boundedness of $\varepsilon - e_\infty$. Conversely, if $\varepsilon \leq e_\infty + K$ on $\mathcal{A}$, then the upper half of the estimate holds with $D - \min(e_\infty) \leq 2d + K$, and the conjecture reduces to the lower half $d \leq C(D - \min(e_\infty)) + c$.

Proof. Take $y = x'$: $d(x, x') = 1$, $D(x, x') = \max(\rho(x, x'), \rho(x', x)) = \rho(x, x') = e_1^*(x) + 1$ (as $x \leq_T x'$), and $e_\infty(x') = e_\infty(x) + 1$, so $d_{\mathcal{T}}(\Phi^*x, \Phi^*x') = 2 \max(e_\infty(x) + 1, e_1^*(x) + 1) - 2e_\infty(x) - 1 \geq 2(e_1^*(x) - e_\infty(x)) + 1$. For the converse, Theorem 1(2) gives $D \leq d + \max(\varepsilon(x), \varepsilon(y)) \leq d + \max(u, v) + K \leq 2d + \min(u, v) + K$ using $|u - v| \leq d$. ■

7.3 Hidden information is unbounded

By Lemma 7 and Proposition 9, everything turns on whether a point can have large first-jump excess and small horofunction level: content in x' that neither x nor many rungs of canonical content supply, at a point whose canonical ascent stays beside the trunk. We call such content *hidden*. The first result says that randomness cannot hide anything for more than two rungs; the second, that genericity hides arbitrarily much.

Lemma 8 (Random sets are dominated one jump up). Let Z be a set and X be Martin-Löf random relative to Z' . Then every total $X \oplus Z$ -computable function is dominated by a Z' -computable function.

Proof. Fix e . For each n the class $\{Y : \Phi_e^{Y \oplus Z}(n) \downarrow\}$ is $\Sigma_1^0(Z)$, so its measure is left-c.e. in Z and Z' computes it to any precision; the measure of $\{Y : \Phi_e^{Y \oplus Z}(n) \downarrow \text{ by stage } s\}$ is Z -computable. Hence Z' computes $h(n) =$ the least s such that $\mu\{Y : \Phi_e^{Y \oplus Z}(n) \text{ halts after stage } s\} < 2^{-n}$. The classes $V_n = \{Y :$

$\Phi_e^{Y \oplus Z}(n)$ halts after stage $h(n)$ are $\Sigma_1^0(Z)$ with indices uniformly Z' -computable and $\sum_n \mu(V_n) < \infty$: a Solovay test relative to Z' . So $X \in V_n$ for only finitely many n , and if $\Phi_e^{X \oplus Z}$ is total then (with the convention that a computation's output is bounded by its halting stage) $\Phi_e^{X \oplus Z}(n) \leq h(n)$ for almost all n . ■

Corollary 4 (No deep hiding with random content). *Let a be 4-random and $g \leq_T a \oplus \emptyset'$ with $a \leq_T g'$. Then $a \leq_T g \oplus \emptyset^{(3)}$ and $e_1^*(g) \leq 2$.*

Proof. A 4-random is generalised-low, so $g' \leq_T (a \oplus \emptyset')' \equiv_T a \oplus \emptyset''$. Since $a \leq_T g'$, a has a g -computable approximation whose modulus m is g' -computable, hence $a \oplus \emptyset''$ -computable. By Lemma 8 with $Z = \emptyset''$ (so $Z' = \emptyset^{(3)}$, relative to which a is random), m is dominated by a $\emptyset^{(3)}$ -computable function f ; then $a(n)$ is the value of the approximation at stage $f(n)$ for almost all n , so $a \leq_T g \oplus \emptyset^{(3)}$. Hence $g' \leq_T a \oplus \emptyset'' \leq_T g \oplus \emptyset^{(3)}$, i.e. $e_1^*(g) \leq 2$. ■

So the random sheet of Section 6, and every construction based on relatively random content, is *transparent* up to two rungs: whatever its jump contains is recovered from the point itself and $\emptyset^{(3)}$. Hidden content needs the opposite property, escape from domination, which generic sets have.

Theorem 4 (Hidden information is unbounded). *For every $E \geq 1$ there is $g \in \mathcal{A}$ with $\varepsilon(g) = e_1^*(g) = E$ and $e_\infty(g) = 0$. Consequently $e_1^* - e_\infty$ and $\varepsilon - e_\infty$ are unbounded on $\mathcal{A}$, and Conjecture 2 is false.*

Proof. Let $a \leq_T \emptyset^{(E+1)}$ be $(E+1)$ -generic [5]. Fix a computable bijection between $\mathbb{N}$ and the disjoint union of a copy of $\mathbb{N}$ (the b -positions) and countably many copies of $\mathbb{N}$ (the columns), and read a as $a = (b, c_0, c_1, \dots)$. Every column is nonempty by genericity; let $m(n) = \min c_n$ and

$$g = \{\langle n, s \rangle : n \in b \wedge m(n) \leq s\}.$$

Then $g \leq_T a$, and b is c.e. in g ($n \in b$ iff some $\langle n, s \rangle \in g$), so $b \leq_T g'$.

Claim: $b \not\leq_T g \oplus \emptyset^{(E)}$. Fix Φ_i . For a string σ (an initial segment of a candidate a) let $\Gamma(\sigma)$ be the partial oracle it determines: $\langle n, s \rangle$ is determined 0 if $\sigma_b(n) = 0$; determined 1 if $\sigma_b(n) = 1$ and σ shows an element of c_n at or below s ; determined 0 if $\sigma_b(n) = 1$ and σ shows $c_n \cap [0, s] = \emptyset$; undetermined otherwise. Let S_i be the set of σ such that for some x there is a halting computation $\Phi_i^{\Gamma(\sigma) \oplus \emptyset^{(E)}}(x)$ using only determined bits, with output different from $\sigma_b(x)$. Membership in S_i is Σ_1 in $\emptyset^{(E)}$, so S_i is Σ_{E+1}^0 , and a meets or avoids it. If a meets S_i then $\Phi_i^{g \oplus \emptyset^{(E)}} \neq b$, since a computation on determined bits of $\Gamma(\sigma)$, $\sigma \subset a$, is a computation on g . Suppose instead $\sigma_0 \subset a$ has no extension in S_i . Choose $x \notin b$ such that neither the b -position x nor any position of column x lies in the domain of σ_0 (b is co-infinite and σ_0 is finite). Suppose $\Phi_i^{g \oplus \emptyset^{(E)}}(x) \downarrow$, and take $\sigma \subset a$, $\sigma \supseteq \sigma_0$, determining every bit used. The output is $0 = b(x)$, else $\sigma \in S_i$. Let σ^{**} be σ with $\sigma_b^{**}(x) = 1$, with every position of column x in the domain of σ set to 0, and extended by zeros on column x up to every s for which $\langle x, s \rangle$ was queried. Then $\sigma^{**} \supseteq \sigma_0$, and every queried bit of $\Gamma(\sigma^{**})$ is determined with the same value as in $\Gamma(\sigma)$: bits $\langle n, s \rangle$ with $n \neq x$ are untouched, and bits $\langle x, s \rangle$ are determined 0 in both ($x \notin b$ in σ ; $x \in b$ with $c_x \cap [0, s] = \emptyset$ in σ^{**}). So the same computation gives $\Phi_i^{\Gamma(\sigma^{**}) \oplus \emptyset^{(E)}}(x) = 0 \neq \sigma_b^{**}(x)$, whence $\sigma^{**} \in S_i$, a contradiction. Hence $\Phi_i^{g \oplus \emptyset^{(E)}}(x) \uparrow$, and again $\Phi_i^{g \oplus \emptyset^{(E)}} \neq b$. This proves the claim.

Now $g' \geq_T b \not\leq_T g \oplus \emptyset^{(E)}$ gives $e_1^*(g) \geq E$, and $g' \leq_T a' \equiv_T a \oplus \emptyset' \leq_T \emptyset^{(E+1)}$ gives $\varepsilon(g) \leq c_1(g) = h(g') - 1 \leq E$; so $\varepsilon(g) = e_1^*(g) = E$ by Lemma 7. For the horofunction level, $g^{(k)} \leq_T a^{(k)} \equiv_T a \oplus \emptyset^{(k)}$ for $k \leq E+1$ (as a is k -generic) and $a^{(k)} \equiv_T \emptyset^{(k)}$ for $k \geq E+1$ (as $a \leq_T \emptyset^{(E+1)}$); and $\emptyset^{(k+1)} \not\leq_T a \oplus \emptyset^{(k)}$ for every k : for $k \leq E$ because a is 1-generic relative to $\emptyset^{(k)}$, so $(a \oplus \emptyset^{(k)})' \equiv_T a \oplus \emptyset^{(k+1)}$, which is impossible if $\emptyset^{(k+1)} \leq_T a \oplus \emptyset^{(k)}$; for $k > E$ trivially. Hence $e_k(g) = 0$ for all k and $e_\infty(g) = 0$. Finally $D(g, g') = e_1^*(g) + 1 = E + 1$ while $d(g, g') = 1$, so Conjecture 2 fails by Proposition 9. ■

The mechanism is worth stating in words. The content b is visible to g in the limit, so one promotion recovers it; but the *time* at which each bit becomes visible is the first element of an independent generic column, revealed only when the bit is 1 and bounded by no oracle short of a' . What is hidden is not the bits but their delays: hidden information is a modulus that escapes domination. The theorem uses $\emptyset^{(E)}$ only through Σ_{E+1} -genericity, so in fact $b \not\leq_T g \oplus z$ for every z relative to which a is 1-generic.

Corollary 5 (Excess is not Lipschitz). *For every E there are $g, y \in \mathcal{A}$ with $d(g, y) = 1$, $\varepsilon(g) = E$ and $\varepsilon(y) = 0$. Hence the excess-level map $\Phi(x) = [x]_{\varepsilon(x)}$ is not a quasi-isometry either.*

Proof sketch. Take g as in the theorem and $y = g \oplus b$. Then $g \leq_T y \leq_T g'$, so $d(g, y) = 1$. The set y is computably equivalent to the pair $(b, m \upharpoonright b)$, and the map $a \mapsto (b, m \upharpoonright b)$ is computable with the property that the image of every basic open set contains a basic open set of the product space $2^{\mathbb{N}} \times \mathbb{N}^b$; so $(b, m \upharpoonright b)$ is $(E+1)$ -generic in that space, hence generalised-low below $\emptyset^{(E+1)}$, and $\varepsilon(y) = 0$. ■

Remark 5 (Status of the sketch). The sketch uses two facts not written out here: that a computable map whose image of every basic open set contains a basic open set carries n -generics to n -generics, and that an n -generic element of a product of Cantor and Baire spaces lying below $\emptyset^{(n)}$ satisfies $y^{(k)} \equiv_T y \oplus \emptyset^{(k)}$ for all k . Both are routine relativisations of the arguments in [5], but the corollary should be read as depending on them. Theorem 4, and the geometric consequences drawn from it in Section 7.4, do not.

The two results together are a dichotomy by growth. Say x' is $\emptyset^{(k)}$ -dominated if every x' -computable total function is dominated by a $\emptyset^{(k)}$ -computable one, and let $\text{dom}(x')$ be the least such k .

Proposition 10 (Excess is bounded by growth). *$e_1^*(x) + 1 \leq \text{dom}(x')$ for every x , and equality holds for the points g of Theorem 4.*

Proof. x' is c.e. in x with an x' -computable modulus; if a $\emptyset^{(k)}$ -computable f dominates it then $x' \leq_T x \oplus f \leq_T x \oplus \emptyset^{(k)}$, so $e_1^*(x) \leq k - 1$. For g : the function $\tilde{m}(n) = m(n)$ if $n \in b$, 0 otherwise, is g' -computable, and a $\emptyset^{(E)}$ -computable function dominating it would give $b \leq_T g \oplus \emptyset^{(E)}$; so $\text{dom}(g') \geq E + 1 = e_1^*(g) + 1$. ■

So first-jump excess is at most the number of rungs needed to dominate the jump, and the bound is attained. Random content is dominated one rung up (Lemma 8) and so hides nothing beyond two rungs; generic content escapes domination at every rung and hides arbitrarily much. Whether ε itself (rather than e_1^*) is controlled by a growth invariant is open.

What remains of tree-likeness is this. The shadow tree is an exact description of $(\mathcal{A}, d)$ on the generalised-low degrees, a coarse one on degrees built from random content, and wrong by arbitrarily much on degrees built from generic content; neither level map into it is a quasi-isometry, and d is not a graph metric (Theorem 3). Conjecture 1 as a bare existence statement is open, but any quasi-isometry to a tree would have to be built from data other than the balls of D ; Section 7.4 reduces it to a bottleneck property of the unit-step graph.

7.4 The unit-step graph is the geodesic model: two flows, minimax reconstruction, one end

The results so far leave $(\mathcal{A}, d)$ looking like a metric with a tree beside it and some puzzling failures between them. This subsection, which follows an external reading of an earlier version of this paper, replaces that picture by a structural one. The rung distance is the interleaving distance of the jump flow; the shadow ultrametric is the interleaving distance of a second, idempotent flow; the excess is the uniform defect between the two; the unit-step graph Γ is a geodesic model of the same coarse geometry, 3-bi-Lipschitz to d ; the shadow tree is recovered from Γ as its minimax edge-weight metric; and the geometry has exactly one coarse end. What remains open is then a single question about Γ .

Theorem 5 (Geodesification). *For all $x, y \in \mathcal{A}$, $d(x, y) \leq d_\Gamma(x, y) \leq 3d(x, y)$. Hence $(\mathcal{A}, d)$ is quasi-isometric to the geodesic graph (Γ, d_Γ) , and Conjecture 1 holds if and only if Γ is a quasi-tree.*

Proof. The left inequality is the triangle inequality along a path. For the right, put $n = d(x, y)$, so $x \leq_T y^{(n)}$ and $y \leq_T x^{(n)}$, and set $z_i = x^{(i)} \oplus y^{(n-i)}$ for $0 \leq i \leq n$. Then $z_0 \equiv_T y^{(n)}$ and $z_n \equiv_T x^{(n)}$. Consecutive terms are within one jump of each other: $z_{i+1} \leq_T z'_i$ since z'_i computes $x^{(i+1)}$ and z_i computes $y^{(n-i-1)}$; and $z_i \leq_T z'_{i+1}$ since z'_{i+1} computes $y^{(n-i)}$ and z_{i+1} computes $x^{(i)}$. So $d(z_i, z_{i+1}) \leq 1$. The canonical ascent $x, x', \dots, x^{(n)} = z_n$, the chain $z_n, \dots, z_0$, and the canonical descent $z_0 = y^{(n)}, \dots, y$ together give a walk of at most $3n$ unit steps. The last clause is Manning's bottleneck criterion [6]: a geodesic space is quasi-isometric to a tree if and only if there is Δ such that every path between the endpoints of a geodesic passes within Δ of its midpoint. ■

Proposition 11 (Two flows). Let $J_k(x) = x^{(k)}$ and $K_k(x) = x \oplus \emptyset^{(k)}$. Then $J_j J_k = J_{j+k}$ is an action of $(\mathbb{N}, +)$ on the degrees and $K_j K_k \equiv_T K_{\max(j,k)}$ an action of the idempotent monoid $(\mathbb{N}, \max)$, and

$$d(x, y) = \min\{k : x \leq_T J_k(y) \wedge y \leq_T J_k(x)\}, \quad D(x, y) = \min\{k : x \leq_T K_k(y) \wedge y \leq_T K_k(x)\}.$$

So d and D are the interleaving distances [7] of the jump flow and of the canonical flow respectively. Jump information accumulates additively and the first geometry supports genuine vertical travel; canonical information accumulates idempotently, which is why the second is non-Archimedean and its balls nest. The excess measures the defect between the flows: with $\eta_k(x) = \min\{e : J_k(x) \leq_T K_{k+e}(x)\}$ (so $\eta_1 = e_1^*$ and $\varepsilon = \sup_k \eta_k$),

$$\varepsilon(x) = \sup_{y \in \mathcal{A}} (\rho(x, y) - \delta(x, y)) :$$

excess is the greatest amount by which the shadow geometry overestimates the directed jump cost from x .

Proof. The two formulas are the definitions unwound. For the last: $\rho \geq \delta$ since $K_k(x) \leq_T J_k(x)$. If $\delta(x, y) = n$ then $y \leq_T x^{(n)} \leq_T x \oplus \emptyset^{(n+\eta_n(x))}$, so $\rho(x, y) - \delta(x, y) \leq \eta_n(x) \leq \varepsilon(x)$. Conversely $\delta(x, x^{(k)}) = k$ and $\rho(x, x^{(k)}) = k + \eta_k(x)$ (a smaller j would give $x^{(k)} \leq_T x \oplus \emptyset^{(j)} \leq_T x^{(j)}$), so the supremum is attained along the canonical ascent. ■

Theorem 6 (The shadow tree is the minimax metric of the unit-step graph). Weight each edge uv of Γ by $D(u, v)$ and let $W(P) = \max_i D(z_i, z_{i+1})$ for a path $P = (z_0, \dots, z_m)$. Then

$$D(x, y) = \min_{P: x \rightsquigarrow y} W(P), \quad \text{and} \quad d_\Gamma(x, y) \leq 2D(x, y) - 1 \quad (x \neq y).$$

Consequently $x E_k y$ (that is, $D(x, y) \leq k$) if and only if x and y lie in the same connected component of the subgraph $\Gamma_{\leq k}$ of edges of weight at most k : the vertices of the ball tree are the components of the unit-step graph filtered by shadow cost.

Proof. Since D is an ultrametric, $D(x, y) \leq W(P)$ for every path. For the converse put $K = D(x, y)$, $x_i = x \oplus \emptyset^{(i)}$ and $y_i = y \oplus \emptyset^{(i)}$. Each $d(x_i, x_{i+1}) \leq 1$ and $D(x_i, x_{i+1}) \leq i + 1$; and x_{K-1}, y_{K-1} are within one jump of each other, since $x'_{K-1} \geq_T x \oplus \emptyset^{(K)} \equiv_T y \oplus \emptyset^{(K)} \geq_T y_{K-1}$ and symmetrically, with $D(x_{K-1}, y_{K-1}) \leq K$. The walk $x_0, \dots, x_{K-1}, y_{K-1}, \dots, y_0$ has at most $2K - 1$ steps and every edge of weight at most K . ■

Corollary 6 (Wormhole edges). For every E there is an edge gy of Γ with $D(g, y) = E + 1$: a single unit step crossing $E + 1$ levels of the shadow tree.

Proof. Take g, b from Theorem 4 and $y = g \oplus b$. Then $g \leq_T y \leq_T g'$, so $d(g, y) = 1$; $\rho(y, g) = 0$; and $\rho(g, y) = \min\{k : b \leq_T g \oplus \emptyset^{(k)}\} = E + 1$, since $b \not\leq_T g \oplus \emptyset^{(E)}$ while $b \leq_T g' \leq_T g \oplus \emptyset^{(E+1)}$. This uses only Theorem 4, not Corollary 5. ■

So hidden information is a geometric fact about Γ : the unit-step graph contains edges that are arbitrarily non-local in the shadow tree. Theorem 6 says the wormholes do not destroy the tree; they are exactly what its edge weights measure. The two geometries are one structure seen with two cost functions.

Proposition 12 (The serial companion metric). $p(x, y) = \delta(x, y) + \delta(y, x)$ is a metric with $d \leq p \leq 2d$. On the random sheet of Section 6, with $M = \max(H, s, t)$,

$$p(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = (M - s) + (M - t),$$

which is exactly the path distance through the meet in the ball tree.

Proof. The directed triangle inequality $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$ gives the triangle inequality for p ; the bounds are $\max \leq \text{sum} \leq 2 \max$. On the sheet, $\delta(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = \max(0, \max(H, t) - s)$ by the computation in Proposition 5 ($b \leq_T a \oplus \emptyset^{(j)}$ iff $j \geq H$, and $\emptyset^{(t)} \leq_T a \oplus \emptyset^{(j)}$ iff $j \geq t$), and symmetrically; the two legs sum to $(M - s) + (M - t)$ in each of the three cases $s, t \leq H$; $s > H \geq t$; $s, t > H$. ■

This settles how Theorem 2 should be read. d is the parallel cost, the longer of the two climbs; p is the serial round-trip, their sum. The random-sheet quadruple that breaks the four-point condition for d is 0-hyperbolic for p , and p is 2-bi-Lipschitz to d . So the four-point failure is the difference between L^∞ and L^1 reconciliation, a coarse-benign artefact of the non-geodesic metric; it says nothing about whether Γ is hyperbolic, and the sheet itself is coarsely a tree. The genuine obstruction to the *specific* shadow tree is the unbounded wormhole phenomenon of Corollary 6, and whether some tree works is the bottleneck question for Γ .

Proposition 13 (Balls, bornology, ends). **1.** $B_d(\emptyset, n) = \{x : x \leq_T \emptyset^{(n)}\}$, so a set is metrically bounded iff its heights are bounded; $\text{diam } B_d(\emptyset, n) = n$ for $n \geq 1$; $(\mathcal{A}, d)$ is countable, integer-valued, uniformly discrete, complete, and unbounded, and the trunk is a geodesic ray, not a Cauchy sequence, so $\emptyset^{(\omega)}$ is an end, not a point of any completion.

2. $\Gamma \setminus B_d(\emptyset, R)$ is connected for every R : the geometry has exactly one coarse end.

Proof. (1) $d(x, \emptyset) = h(x)$; the diameter bound is Lemma 2 and Lemma 3 ($d(a, \emptyset^{(H)}) = H$). (2) For x with $h(x) = H > R$, the path $x \oplus \emptyset^{(k)}$, $k = 0, \dots, H$, has unit steps, stays at height H , and ends at $\emptyset^{(H)}$; two points outside the ball are joined through the trunk between $\emptyset^{(h(x))}$ and $\emptyset^{(h(y))}$ without entering it. ■

One coarse end is a statement about Γ at large scale; the ray boundary $\partial_\infty \mathcal{A}$ of Proposition 6 is finer, since several asymptote classes of geodesic rays can lie in one coarse end, and whether it is a singleton remains open.

What is now known about the geometry can be put in one line each: d is the jump-interleaving metric; D is the canonical-flow interleaving ultrametric; ε is their uniform defect; Γ is the geodesification, with $d \leq d_\Gamma \leq \min(3d, 2D - 1)$; D is the minimax edge-weight metric of Γ and its balls are the filtered components; unit edges cross arbitrarily many tree levels; and there is one coarse end. What is open is whether Γ has the bottleneck property (expected not; Section 7.5), whether the ray boundary is a singleton, and the exact value of d_Γ on the sheet.

7.5 Finite metric universality: a conditional refutation of tree-likeness

The bottleneck question for Γ can be attacked through the embedding theory of the jump, and an external reading of an earlier version pointed out that it may be closed negatively that way. The argument is short and we give it in full; its one non-elementary step is a hypothesis about countable jump partial orders that, as far as we can determine, is open in the literature, so the conclusion is stated as conditional.

A *jump partial order with least element* is a structure $(P, \leq, ', 0)$ with 0 least, $x < x'$, and $x \leq y \Rightarrow x' \leq y'$. The Turing degrees $(\mathcal{D}, \leq_T, ', \mathbf{0})$ are one. Hinman and Slaman [8] proved that every countable jump partial order embeds in $\mathcal{D}$ preserving $\leq$ and $'$; Montalbán [9] extended this to jump upper semilattices, and showed that with 0 in the language the semilattice version fails (some one-generator types over $\mathbf{0}$ are not realised), while every one-generator type of jump *partial order* over $\mathbf{0}$ is realised. The case of several generators over $\mathbf{0}$ is, in the sources we have found, posed as open.

Hypothesis (H). Every countable jump partial order with least element embeds in $(\mathcal{D}, \leq_T, ', \mathbf{0})$ preserving $\leq$, non- $\leq$, $'$ and 0. (A jump partial order is infinite whenever it is nonempty, since $x < x'$; what (H) adds to Hinman–Slaman is the clause on 0.)

Theorem 7 (Finite metric universality, under (H)). *Assume (H). Then every finite metric space (F, μ) with integer distances embeds isometrically in $(\mathcal{A}, d)$. Consequently Γ contains isometrically embedded cycles of every length, Γ is not Gromov-hyperbolic, and $(\mathcal{A}, d)$ is not quasi-isometric to any tree: Conjecture 1 is false.*

Proof. Let $N = \text{diam}(F) \geq 1$. Define a structure on the symbols $x_{v,r}$ ($v \in F, r \in \mathbb{N}$) and 0_r ($r \in \mathbb{N}$) by $x'_{v,r} = x_{v,r+1}$, $0'_r = 0_{r+1}$, and

$$\begin{aligned} x_{v,r} \leq x_{w,s} &\iff r + \mu(v, w) \leq s, & 0_r \leq x_{v,s} &\iff r \leq s, \\ x_{v,r} \leq 0_s &\iff r + N \leq s, & 0_r \leq 0_s &\iff r \leq s. \end{aligned}$$

This is a partial order: antisymmetry uses $\mu(v, w) > 0$ for $v \neq w$ and $N \geq 1$; transitivity is the triangle inequality for μ in the case $x \leq x \leq x$, and $\mu \leq N$ in the cases through 0 (for instance $x_{v,r} \leq 0_s \leq x_{w,t}$ gives $r + \mu(v, w) \leq r + N \leq s \leq t$). The jump is strictly increasing and monotone, since each defining inequality is preserved by adding 1 to both indices, and 0_0 is least. It is countable, so by (H) it embeds in $\mathcal{D}$ with $0_r \mapsto \emptyset^{(r)}$ and $x_{v,r} \mapsto a_v^{(r)}$ for degrees a_v ; only the indices $r \leq 2N$ are used below. Then $a_v \leq_T \emptyset^{(N)}$, so $a_v \in \mathcal{A}$, and for $k \leq N$, $a_w \leq_T a_v^{(k)}$ iff $\mu(v, w) \leq k$; hence $\delta(a_v, a_w) = \mu(v, w)$ and, μ being symmetric, $d(a_v, a_w) = \mu(v, w)$.

For the cycle C_n with its path metric, the embedded vertices $a_0, \dots, a_{n-1}$ have consecutive ones adjacent in Γ , so the shorter arc between a_i and a_j is a Γ -path of length $\mu(i, j)$; with $d \leq d_\Gamma$ this gives $d_\Gamma(a_i, a_j) = \mu(i, j)$, an isometric cycle in Γ . On C_{4m} the four quarter points have d_Γ -sums $4m, 2m, 2m$, so the four-point condition forces $\delta \geq m$; as m is arbitrary, the geodesic space Γ is not hyperbolic, hence not a quasi-tree, and by Theorem 5 neither is $(\mathcal{A}, d)$. ■

Three remarks. First, the hypothesis is used only for the specific jump partial orders built from cycles, so what is needed is less than (H): an arithmetical realisation of a cyclic chain $a_0, \dots, a_{n-1} \leq_T \emptyset^{(N)}$ with $a_{i\pm 1} \leq_T a_i'$ and $a_j \not\leq_T a_i^{(k)}$ for $k < \mu(i, j)$. Second, the published results do not give this: without the clause on 0 the Hinman-Slaman embedding places the configuration in the cone above some degree z , where it shows that that component of $(\mathcal{D}, d)$ contains isometric cycles and is not coarsely a tree, which bears on open problem 4. but not directly on $\mathcal{A}$. Third, the natural route to an unconditional proof is the one used for Theorem 4: realise all a_i as computable images of a single sufficiently generic set, with each $a_{i\pm 1}$ c.e. in a_i through delays drawn from private columns, so that the positive relations hold by construction and the negative ones by the density argument of that proof. We have not carried this out.

8 OPEN PROBLEMS

1. **[Open] The graph Γ .** d is not a graph metric (Theorem 3) but $d \leq d_\Gamma \leq 3d$ (Theorem 5). What is $d_\Gamma(a, b)$ exactly on the random sheet? Corollary 3 gives $[3t - 1, 4t - 1]$ by applying the no-common-information lemmas at a single index; applying them along the whole path should sharpen the lower bound, and if $d_\Gamma(a, b) = 4t - 1$ then off-sheet points never help. Is Γ Gromov-hyperbolic? Theorem 2 does not decide this, since d is not geodesic.
2. **[Open] Tree-likeness.** Both level maps into the shadow tree fail: the horofunction-level map by Theorem 4, the excess-level map by Corollary 5. Is $(\mathcal{A}, d)$ quasi-isometric to any tree, that is, does Γ have the bottleneck property? Under Hypothesis (H) of Section 7.5 it is not; an unconditional proof needs isometric cycles in $\mathcal{A}$, for instance by the generic-coding construction sketched at the end of that section, or a published realisation theorem for jump partial orders with 0 in several generators. Whether such cycles can be built with $e_\infty(a_i) = 0$, so that they live on the horosphere level 0 and the four-point failure is not a highness effect, is a further question. If it were, the tree would not be the ball tree of D (Corollary 6). Is ε (as opposed to e_1^*) bounded by a growth invariant?
3. **[Open] The boundary.** Is $\partial_\infty \mathcal{A} = \{\xi\}$? By Proposition 6(3), this is the question whether every geodesic ray has bounded lag behind the trunk. A ray of points each random over the trunk at its own level cannot have consecutive generalised-low points with $e_k = 0$ one jump apart (such points would be E_1 -equivalent, bounding all heights), so any counterexample must use points of positive excess at every step; whether the points of Theorem 4 can be strung along a ray is the natural next attempt.
4. **[Open] Other components.** d is finite on each class of mutually arithmetical degrees. Whether the components other than $\mathcal{A}$ (those of $\emptyset^{(w)}$, of a non-arithmetical random, of a hyperarithmetical H_a) are isometric, quasi-isometric, or geometrically distinct from $\mathcal{A}$ is not examined. Relativisation gives an isometric copy of $\mathcal{A}$ inside each component, but a component may be larger than that copy.
5. **[Open] Excess as an invariant.** $\varepsilon(x) = 0$ exactly on the generalised-low degrees. The distribution of excess (which values occur on which classes; whether $\varepsilon(x \oplus y)$ is controlled by $\varepsilon(x), \varepsilon(y)$; the excess of n -generics) is not worked out.

6. [Open] Facts relied upon, and related work. Lemma 3 relies on the relativised low basis theorem, van Lambalgen’s theorem and the Kautz jump fact, each in relativised form; these are standard, and the paper uses them in exactly the forms stated there. A systematic comparison with the literature is also outstanding: the author is not aware of a metric or coarse structure previously placed on a degree structure, the nearest things being the jump hierarchy itself and the n -generic and n -random “distance from the trunk” facts of Jockusch, Kautz and Kurtz; the abstract form of d and D as interleaving distances of flows on a poset is standard [7], and what is claimed as new is the geometry of this particular pair of flows; Lemma 8 is presumably known in some form (the case $Z = \emptyset$ says that 2-randoms are $\emptyset'$ -dominated); and Lemmas 5 and 6 are in the spirit of the Downey–Nies–Weber–Yu characterisation of weak 2-randomness by Π_2^0 null sets [16], of which they may be special cases.

9 CONCLUSION

The rung distance turns the jump hierarchy into a geometry with one vertical direction, one ideal point and one coarse end, and this paper has established what that geometry is at the level of coarse invariants: canonical ascents are geodesic rays converging to a single point at infinity whose Busemann function is the horofunction level of highness; the shadow ultrametric D and its ball tree describe the space exactly on the generalised-low degrees and up to excess elsewhere; d is neither Gromov-hyperbolic in the four-point sense nor a graph metric, but is 3-bi-Lipschitz to the unit-step graph Γ , of which D is the minimax edge-weight metric; hidden information, the modulus that a jump can conceal, is unbounded, so neither level map into the ball tree is a quasi-isometry; and tree-likeness is reduced to Manning’s bottleneck property for Γ and refuted under Hypothesis (H). What remains open is listed in Section 8.

The thread continues in the companion papers. The core paper [2] summarises the present results as the second of its three orderings, geometry beside reachability and certification [2, Sections 7 and 13]; Paper C [3] puts the geometry on a clock, where the boundary question of Section 8 becomes the question whether a population has one horizon or several, and uses the hidden-information results of Section 7.3 to separate what an observer can predict about another agent from what it can audit in hindsight [3, Section 5]; Paper D [4] takes the resulting rate dynamics to the Fermi problem. The distinction that runs through all four, between a limit that exists, an approximation that converges to it, and a modulus that certifies the convergence, is the one Theorem 4 makes geometric: what a degree can hide is precisely such a modulus.

AUTHOR CONTRIBUTIONS

The rung distance and the programme of studying its geometry, the identification of ξ with true arithmetic, and the direction of the argument are due to the author. The statements and proofs (the Shadow Tree Theorem, the closed form on the random sheet, the non-hyperbolicity theorem, the ray proposition, the reduction of the tree-likeness conjecture, the dichotomy, the dead-end theorem with its two no-common-information lemmas, the excess sandwich, the hidden-information theorem and its corollaries, the geodesification, two-flow and minimax results, and the conditional universality theorem), the identification of the relevant literature and much of the drafting were developed with Claude, an AI system developed by Anthropic, used under the author’s direction in extended dialogue. Claude is not listed as an author because it cannot take responsibility for the work, respond to review or correct the record. The author takes full responsibility for all claims, proofs and errors. The proofs were developed with an AI system and reviewed by the author; they have not been independently refereed.

REFERENCES

- [1] N. Goodman. Rungs, Voids and Effective Closure: a programme synopsis. Working paper (Paper O), 2026.
- [2] N. Goodman. Rungs, voids and effective closure: cognitive spaces, uncertified convergence and promotion. Working paper (Paper B), 2026.
- [3] N. Goodman. Populations and strategy in cognitive spaces: lag, horizons, finite computation and the Gambler. Working paper (Paper C), 2026.

- [4] N. Goodman. Cognitive accessibility and the Fermi problem: a rate model, a Fermi function and a marked birth-and-contact process. Working paper (Paper D), 2026.
- [5] C. G. Jockusch. Degrees of generic sets. In *Recursion Theory: its Generalisations and Applications*, London Math. Soc. Lecture Note Ser. 45, Cambridge, 1980, 110–139.
- [6] J. F. Manning. Geometry of pseudocharacters. *Geom. Topol.* 9 (2005), 1147–1185.
- [7] V. de Silva, E. Munch and A. Stefanou. Theory of interleavings on categories with a flow. *Theory Appl. Categ.* 33 (2018), 583–607; P. Bubenik, V. de Silva and J. Scott. Metrics for generalized persistence modules. *Found. Comput. Math.* 15 (2015), 1501–1531.
- [8] P. G. Hinman and T. A. Slaman. Jump embeddings in the Turing degrees. *J. Symbolic Logic* 56 (1991), 563–591.
- [9] A. Montalbán. Embedding jump upper semilattices into the Turing degrees. *J. Symbolic Logic* 68 (2003), 989–1014.
- [10] R. M. Friedberg. A criterion for completeness of degrees of unsolvability. *J. Symbolic Logic* 22(2), 1957.
- [11] D. B. Posner and R. W. Robinson. Degrees joining to $\mathbf{0}'$. *J. Symbolic Logic* 46(4), 1981.
- [12] C. G. Jockusch and R. I. Soare. Π_1^0 classes and degrees of theories. *Trans. Amer. Math. Soc.* 173, 1972.
- [13] S. M. Kautz. *Degrees of Random Sets*. PhD thesis, Cornell University, 1991.
- [14] E. L. Post. Degrees of recursive unsolvability (abstract). *Bull. Amer. Math. Soc.* 54, 1948.
- [15] G. E. Sacks. *Degrees of Unsolvability*. Annals of Mathematics Studies 55, Princeton, 1963.
- [16] R. Downey, A. Nies, R. Weber and L. Yu. Lowness and Π_2^0 nullsets. *J. Symbolic Logic* 71(3), 2006.
- [17] J. R. Shoenfield. On degrees of unsolvability. *Ann. of Math.* 69, 1959.
- [18] R. I. Soare. *Turing Computability: Theory and Applications*. Springer, 2016.
- [19] A. Tarski. Der Wahrheitsbegriff in den formalisierten Sprachen. *Studia Philos.* 1, 1936.
- [20] M. van Lambalgen. The axiomatization of randomness. *J. Symbolic Logic* 55(3), 1990.

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